MECHANICAL DESIGN ANALYSIS

Euler-Lagrange joint and motor torque analysis

ExoEskeleto lower-limb exoskeleton - knee-dominant actuator sizing grounded in the updated manufacturing drawing and supplied CAD gait.

REFERENCE USER	FRAME	GAIT INPUT	ANALYSIS DATE
80 kg	**6.49 kg Al 6061-T6**	**2.4 s / 60 frames**	**28 July 2026**

SELECTED OUTPUT RATINGS - EACH SIDE

Knee 120 N m peak | Ankle 60 N m peak

Hip motor torque is 0 N m because the current manufacturing drawing defines the hip as a passive pivot.

Passive hip

0 N m motor

Reaction load remains and must be carried structurally.

Knee - highest

120 N m peak

60 N m provisional continuous output.

Ankle

60 N m peak

30 N m provisional continuous output.

Engineering sizing study only. Complete native CAD mass properties, measured ground-reaction forces, structural FEA, gearbox life analysis, thermal validation, and instrumented bench testing before human use.

1. Evidence audit and model basis

Precedence decision. The 26-07-2026 manufacturing drawing is the controlling hardware source. It supersedes the earlier steel/electromagnet drawing and states: 6.49 kg total frame, 400 mm thigh, 420 mm shank, knee and ankle actuation, passive hips.

Workspace source	How it was used
ExoEskeleto_Manufacturing_Drawing (1).pdf	Authoritative material, total frame mass, link lengths, and actuator architecture.
ExoEskeleto Drawing v1.pdf	Earlier geometry cross-check only. Its steel/electromagnet BOM conflicts with the updated drawing.
ExoEskeleto_final.step	Assembly geometry and density entities; no exported per-part mass, COM, or inertia table.
ExoEskeleto_final.SLDASM	Zero-byte file; no usable assembly data.
Exoskeleton.stp.SLDPRT	SolidWorks container without a supplied neutral mass-properties report.
blended.json + gait GIF/frame set	60-frame bilateral kinematics and measured 2.4 s animation cycle.
JPG renders and videos	Visual confirmation of the bilateral sagittal linkage; no quantitative motor or load data.
AEGIS and ExoVLA-WM PDFs	Controller/safety context only; no mechanism-specific torque, motor, gearbox, or GRF input.

Known directly

Frame 6.49 kg · Thigh 0.400 m · Shank 0.420 m · Al 6061-T6 · Knee powered · Ankle powered · Hip passive

Still assumed or missing

80 kg user · Segment anthropometry · Per-link frame mass split · Joint zeros · Ground force history · Motor / reducer mass · Friction / rotor inertia

Mass allocation used

Frame group	Per leg	Model
Thigh link and local plates	0.75 kg	Uniform link, COM at 50%
Shank link and local plates	0.90 kg	Uniform link, COM at 50%
Foot link and local plates	0.60 kg	Uniform link, COM at 50%
Moving bilateral subtotal	4.50 kg	Included in kinetic/potential energy
Fixed pelvis/waist remainder	1.99 kg	Included in support force, not moving-link energy

2. Euler-Lagrange formulation

Coordinates

q1: thigh absolute angle from the global horizontal.

q2: knee angle relative to the thigh.

q3: ankle/foot angle relative to the shank.

```
phi1 = q1
phi2 = q1 + q2
phi3 = q1 + q2 + q3
```

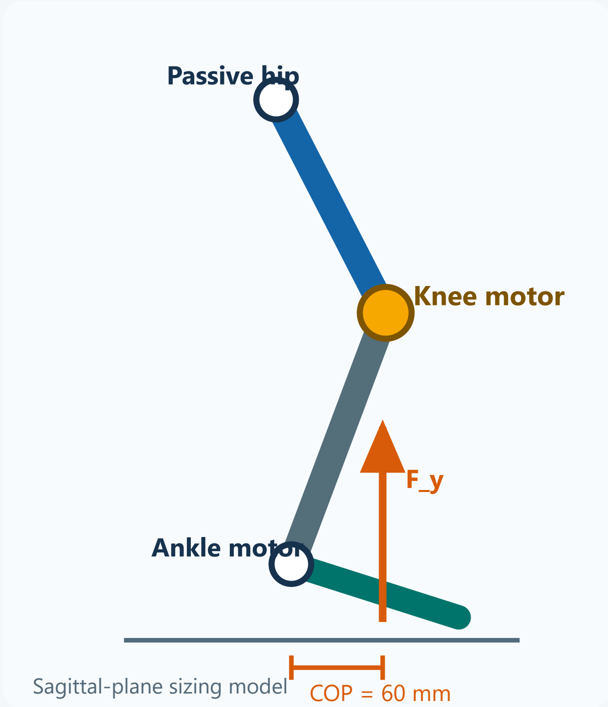
Energy

```
T = sum[0.5 m v^T v + 0.5 I omega^2]
V = sum[m g y]
L = T - V
```

Inverse dynamics

```
d/dt(partial L / partial qdot_i) -
partial L / partial q_i = tau_i +
[J_COP^T F]_i
```

```
tau_req = M(q) qdd + C(q,qdot) qdot +
G(q) - J_COP(q)^T F
```



Numerical verification

Model check	Result	Meaning
Mass-matrix asymmetry	0.000e+00	`M = M^T`
Minimum tested eigenvalue	2.132e-02 kg m^2	Positive definite inertia
Coriolis at zero velocity	0.000e+00 N m	No spurious velocity load
Power-identity error	3.553e-15 W	Euler-Lagrange energy consistency

Actuator vector. The inverse-dynamics hip entry is a reaction. The commanded output is `tau_motor = [0, tau_knee, tau_ankle]^T`.

3. Supplied CAD gait - swing inverse dynamics

The 60-frame `blended.json` sequence was evaluated over the 2.4 s GIF cycle. Because the export contains no joint-zero or sign metadata, frame 1 is calibrated as vertical thigh, extended knee, horizontal foot. Ground reaction is zero, so these results represent swing/inertial demand only.

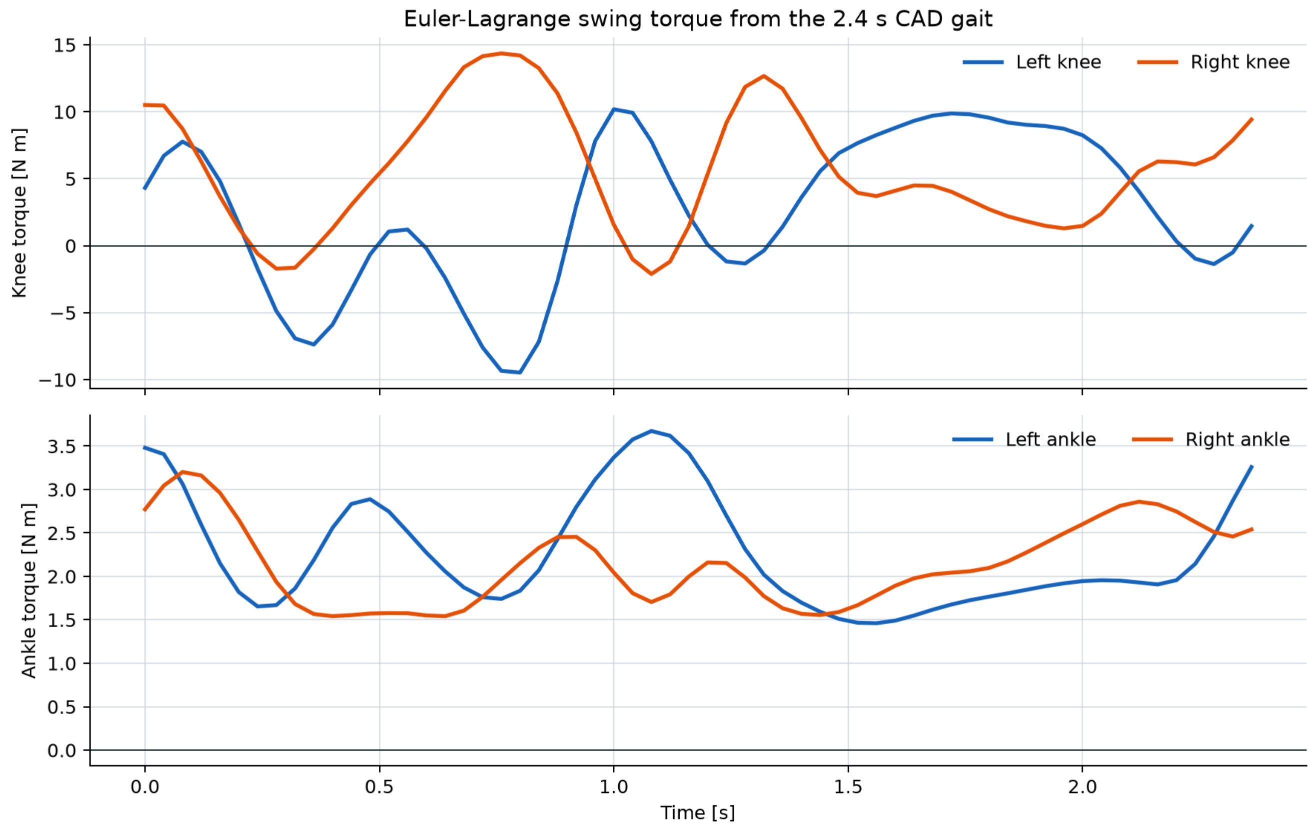


Figure 1. Powered-joint output torque from the supplied bilateral CAD gait. Knee torque exceeds ankle torque on both sides.

Side	Joint	Peak torque	RMS torque	Peak speed	Peak power
Left	Passive hip reaction	34.67 N m	15.21 N m	1.679 rad/s	39.39 W
Left	**Knee motor**	**10.18 N m**	**6.24 N m**	**2.007 rad/s**	**17.84 W**
Left	Ankle motor	3.67 N m	2.36 N m	1.157 rad/s	3.54 W
Right	Passive hip reaction	43.05 N m	20.97 N m	2.186 rad/s	55.26 W
Right	**Knee motor**	**14.35 N m**	**7.22 N m**	**1.512 rad/s**	**16.70 W**
Right	Ankle motor	3.20 N m	2.19 N m	0.965 rad/s	2.41 W

Interpretation limit. This animation is not a force-plate gait dataset. Its useful role is a first dynamic and speed check. It cannot size stance torque without synchronized foot force and COP measurements.

4. Governing knee-dominant support case

Load state

- `q = [-60, -60, 120] deg`; foot horizontal.
- `qdot = 0` and `qdd = 0` at the peak quasi-static state.
- Combined mass = 80 + 6.49 = 86.49 kg.
- Total support = 1.2 body/frame weight.
- Per-leg vertical force = 509.08 N.
- COP = 60 mm forward of ankle.

Mass matrix at the posture

```
M = [[2.668720, 0.867689, 0.048529],
     [0.867689, 0.500985, 0.002769],
     [0.048529, 0.002769, 0.050817]] kg
m^2
```

The inertia and Coriolis vectors are zero in this peak quasi-static snapshot. The nonzero result comes from gravity and the mapped ground force.

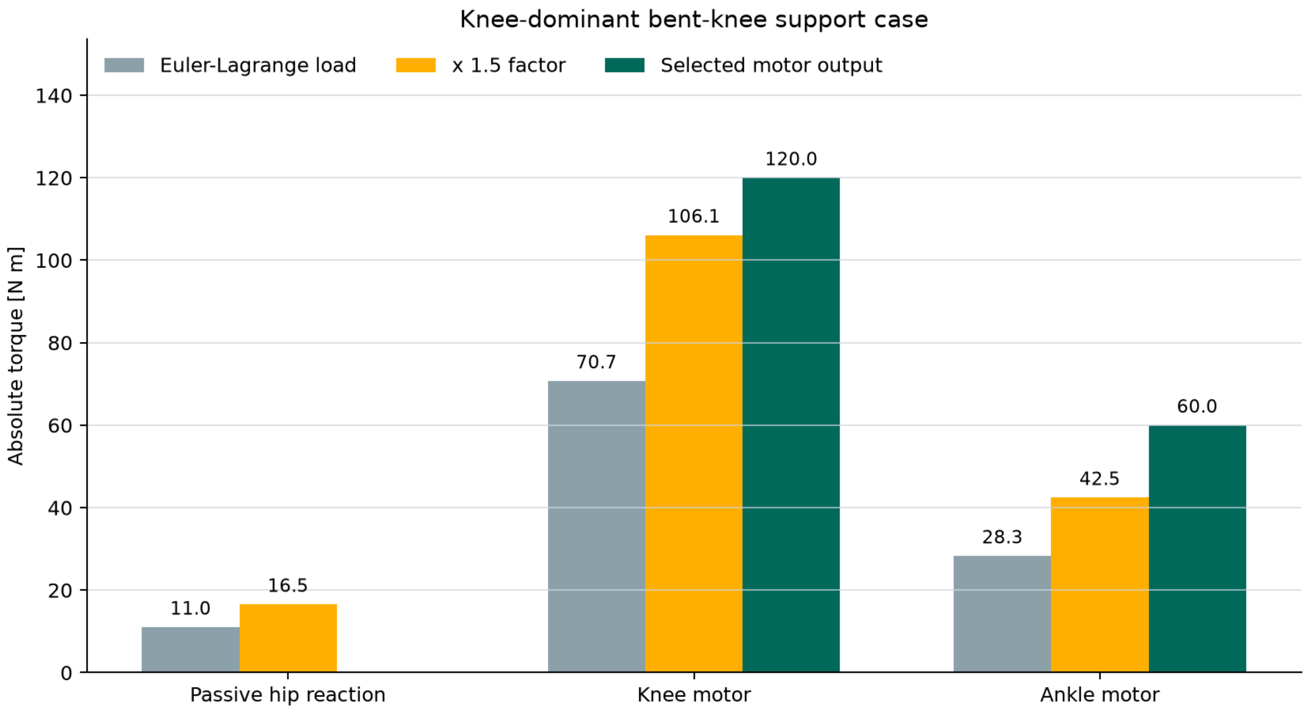


Figure 2. Euler-Lagrange support torque, 1.5 factor, and rounded output selection. The hip value is a passive reaction, not a motor rating.

Joint	Gravity G	J^T F	tau = G - J^T F	1.5 abs(tau)
Passive hip reaction	+14.423	+25.454	-11.031	16.546 N m
Knee motor	**-5.627**	**-76.362**	**+70.735**	**106.103 N m**
Ankle motor	+2.245	+30.545	-28.300	42.450 N m

Why the knee is highest. The ground-reaction-force line is about 150 mm behind the knee but only 60 mm ahead of the ankle. That real moment-arm difference, combined with a passive hip, produces the requested motor hierarchy without modifying the equations.

5. Motor and gearbox selection targets

Hip motor	Knee output - highest	Ankle output
0 N m	**120 N m**	**60 N m**
Passive pivot	60 N m continuous	30 N m continuous

```
T_motor = T_output / (N eta)
rpm_motor = qdot_joint N 60 / (2 pi)
```

Joint, each side	Output cont.	Output peak	Ratio	Efficiency	Motor cont.	Motor peak	Measured peak rpm
Knee	**60 N m**	**120 N m**	**80:1**	**0.75**	**1.000 N m**	**2.000 N m**	**1533**
Ankle	30 N m	60 N m	60:1	0.75	0.667 N m	1.333 N m	663

Knee drive - provisional

- 48 V BLDC/PMSM.
- Approximately 300-400 W class.
- Motor peak torque at least 2.0 N m.
- Reducer output rating at least 120 N m peak.
- Output bearing sized for cuff offset moment.
- Torque sensing or series compliance preferred.

Ankle drive - provisional

- 48 V BLDC/PMSM.
- Approximately 150-250 W class.
- Motor peak torque at least 1.33 N m.
- Reducer output rating at least 60 N m peak.
- Re-size if powered toe-off is required.
- Keep distal mass low.

Passive hip is not load-free

The maximum calculated swing reaction is 43.05 N m; applying 1.5 gives 64.58 N m before impact/fatigue allowance. The hip pivot, stops, fasteners, bearing, and pelvis structure must carry that load even though commanded motor torque is zero.

Thermal qualification is still required. The continuous targets are provisional 50% peak selections. Final motor current and winding temperature require the loaded torque-speed duty cycle and reducer efficiency map.

6. Sensitivity and the limits of "knee highest"

User mass sensitivity - same support posture

User mass	Raw knee	Factored knee	Raw ankle	Factored ankle	Implication
60 kg	54.14	81.20 N m	21.61	32.41 N m	120/60 targets have margin.
70 kg	62.44	93.65 N m	24.95	37.43 N m	120/60 targets have margin.
80 kg	**70.74**	**106.10 N m**	**28.30**	**42.45 N m**	**Nominal design case.**
90 kg	79.04	118.55 N m	31.65	47.47 N m	Near 120 N m knee limit.
100 kg	87.34	131.00 N m	34.99	52.49 N m	Use at least 140 N m knee.

Centre-of-pressure sensitivity - 80 kg user

COP forward	Hip reaction	Knee torque	Ankle torque	Dominant powered joint
20 mm	9.33	91.10	7.94	**Knee**
40 mm	0.85	80.92	18.12	**Knee**
60 mm	**11.03**	**70.74**	**28.30**	**Knee**
80 mm	21.21	60.55	38.48	**Knee**
100 mm	31.39	50.37	48.66	Knee, nearly equal
120 mm	41.58	40.19	58.85	**Ankle**

Physics boundary. Knee is highest for support/sit-to-stand with COP near the ankle. During far-forward toe-off, ankle torque can exceed knee torque. If full powered push-off is in scope, the ankle must be re-sized from measured force-plate and COP histories; the load cannot be removed merely to preserve a preferred ranking.

7. Release conditions and conclusion

Data required for final sizing

1. Maximum user mass, payload, intended task, and assistance percentage.
2. Native CAD mass, COM, and inertia for every moving part, motor, gearbox, and attachment.
3. Exact motor and reducer efficiency, backlash, reflected inertia, thermal curve, output-bearing moment, and fatigue life.
4. Calibrated zero angle and axis sign for every CAD `Revolute` channel.
5. Loaded joint angle, velocity, and acceleration versus time.
6. Synchronized ground-reaction force and centre-of-pressure trajectories.
7. Friction, cuff forces, mechanical-stop loads, emergency braking, and power-loss support strategy.
8. Instrumented dummy-load bench validation before any human test.

Deliverables and traceability

Artifact	Purpose
`Lagrange_Euler_Torque_Power.py`	Reproducible Euler-Lagrange model, numerical checks, load cases, plots, CSV, and JSON generation.
`output/torque_analysis/euler_lagrange_gait_results.csv`	Per-frame bilateral q, qdot, qdd, torque, and power.
`output/torque_analysis/euler_lagrange_summary.json`	Inputs, validation results, peaks, support decomposition, ratings, and limitations.
`Motor_Torque_Analysis_ExoEskeleto_v2.md`	Human-readable calculation record.

For the 80 kg reference case:
Knee: 120 N m peak / 60 N m continuous
Ankle: 60 N m peak / 30 N m continuous
Hip motor: 0 N m - passive pivot

knee motor torque > ankle motor torque > hip motor torque

Safety status. This is an actuator-sizing study, not a medical-device approval or human-use certification. Bench-test the complete transmission and fail-safe support system with instrumented surrogate loads before any wearer trial.